

SILICON VISION

Prof V. Kamakoti's India chip blueprint

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Few voices carry as much moral and technical authority in India's semiconductor journey as Prof V. Kamakoti, Director of IIT Madras and one of the country's most influential champions of sovereign computing.

As the driving force behind the Shakti indigenous processor programme and a key architect of India's academic-industry chip ecosystem, Kamakoti represents the bridge between policy ambition and engineering reality. At a time when India is racing to build fabs, design talent and deep-tech confidence, I was keen on his perspective for the road ahead.

If you had to write India's 'compute sovereignty' doctrine for 2035, what are the non-negotiables?

Prof Kamakoti: For India to achieve compute sovereignty by 2035, we need to achieve the following things:

- Our own foundry that can enable us to manufacture designs at advanced technology nodes, thereby, decreasing our reliance on outsourced manufacturing.
- We need to train the next generation of students to be able to fill the need for hardware engineers.

What's the most misunderstood aspect of India's semiconductor industry?

Prof Kamakoti: We have already put out silicon proven designs through Shakti, Mindgrove, and InCore out in the field. I think people think of indigenous hardware manufacturing as a largely research effort but what we are aiming for is to provide access to industrial class hardware in an open-source manner.

If you could pick just two national 'moonshots' for Indian academia in chips over the next decade?

Prof Kamakoti: Security, and EDA - I think we are in a decent shape when it comes to processor design at the architecture and micro-architecture level, thanks to the dedicated efforts of the students who have worked with us over the years.

Right now, at least for the next five years, the aim is to showcase that we can design and develop the next generation of hardware that can run AI workloads, can incorporate security features, and can support multi-core architectures. We are also seeing promising developments on the indigenous fab, thanks to the strong support from MeITY and the central government. It is now time to

start democratising our efforts on the physical design or the EDA front.

How do we create 10,000 'hardware-native' engineers (not just coders who learned RTL) without diluting quality? What should change in curricula, labs, internships, and faculty development?

Prof Kamakoti: I think it can't be done overnight and this is one area where we all need to work together. The larger IITs have the responsibility to create the next generation of faculty who can then take the mission forward. If we can even motivate eight students a year to buy into this goal and take the responsibility of training the next generation of undergraduates, then in five years we can have a workforce of around 2.5 lakh hardware engineers.

But the success of this mission depends on everyone working together in unison to realize the dream of indigenous semiconductor sovereignty. IIT Madras, under the aegis of the RISC-V Knowledge Center of Excellence (a MeITY funded initiative), has been training students across the country via workshops and events such as the Digital India RISC-V (DIR-V) Symposium. 